

Lecture 4: Conditions for Unbiasedness and Consistency of OLS

In this lecture, we will discuss the conditions we need for OLS to be unbiased and the conditions we need for OLS to be consistent.

Correlation vs. Causation

If you have heard anything about statistics, you have probably heard the phrase “correlation is not causation.” This phrase means that the relationship between variables in the data may not be the same as the causal relationship.

We now have more sophisticated tools to talk about this. We will focus here on the full population, rather than the sample. This highlights that the issue of correlation vs. causation is not something that is solved by getting a larger sample, rather, it is an issue of interpreting coefficients in the population. Suppose that we have the causal model

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

where β reflects the true causal effect of X_i on Y_i , and ε_i includes all other factors besides X_i that affect Y_i .¹ Here, β measures how much Y_i would go up if we increased X_i by one unit, holding everything else constant.

Separate from β , which measures the causal effect, is the best linear predictor, β^{BLP} , which measures the correlational relationship between Y_i and X_i . As we saw last lecture, β^{BLP} is the population counterpart to OLS: it is the value of β that minimizes the squared deviations of Y_i from $\alpha^{BLP} + \beta^{BLP} \cdot X_i$. We showed that is equivalent to requiring that $E[Y_i - \alpha^{BLP} - \beta^{BLP} \cdot X_i] = 0$, and $\text{Cov}(X_i, Y_i - \alpha^{BLP} - \beta^{BLP} \cdot X_i) = 0$. We called these orthogonality conditions, because they reflect that the regressor and the error term must be orthogonal (zero covariance).

When will the causal effect, β , be the same as the correlational relationship, β^{BLP} ? The two effects will be equal only if the orthogonality condition holds for the residuals of the causal model, ε_i . If the causal β did not result in ε_i being orthogonal to X_i , then it could not be β^{BLP} .

As economists, we are mostly interested in causal effects. Empirical economists spend much of our time debating whether the error term is orthogonal to the regressor, as well as devising ways to make sure that is the case. The orthogonality condition will be the most important assumption behind OLS, and you will see it come back in new forms when we study other methods to estimate causal effects.

However, sometimes we are actually interested in prediction, rather than in causal effects. For example, we may be interested in predicting inflation next quarter, given a set of variables we can measure today. Or we may wish to predict who is a good credit risk, based on their previous credit history, or which stocks tend to go up the most when there is a downturn, as a hedge against recessions. For prediction, we are actually interested in β^{BLP} , and so the orthogonality condition will be satisfied automatically.

Exercise 1. A college admissions office is trying to figure out who will do well in their university. They run a regression and find that students who took more AP courses in high school get higher grades in their freshman year. Does this regression identify the causal effect of taking AP courses on college performance? Why might the error term be correlated with the regressor?

Exercise 2. In the exercise above, would the admissions office be most interested in a causal effect or in prediction?

¹Since we have included an intercept, ε_i really measures how much higher these factors are for observation i than they are for the average observation.

Consistency of OLS

We are now ready to lay out the assumptions for OLS to be a consistent estimator of the true effect. Consider the following model:

$$Y_i = \alpha + \sum_{k=1}^K \beta_k \cdot X_{k,i} + \varepsilon_i$$

Note that the number of regressors, K , could equal one, in which case this would be OLS with a single regressor, but it also could have $K > 1$.

Assumptions for Consistency

In order for OLS to be consistent, we need three assumptions.

Assumption (No Perfect Collinearity/Full Rank Condition). *None of the regressors are perfectly collinear: it is not possible to perfectly predict one of the regressors as a linear combination of the other regressors and a constant.*

This condition is necessary for least squares to have a unique solution. Suppose, for example, that $Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \varepsilon_i$, but we had a perfect collinearity such that $X_{1,i} = X_{2,i} + X_{3,i}$. Then, substituting this into the original model, we would have:

$$\begin{aligned} Y_i &= \alpha + \beta_1 (X_{2,i} + X_{3,i}) + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \varepsilon_i \\ &= \alpha + 0 \cdot X_{1,i} + (\beta_1 + \beta_2) X_{2,i} + (\beta_1 + \beta_3) X_{3,i} + \varepsilon_i \\ &= \alpha + \tilde{\beta}_1 \cdot X_{1,i} + \tilde{\beta}_2 X_{2,i} + \tilde{\beta}_3 X_{3,i} + \varepsilon_i \end{aligned}$$

where $\tilde{\beta}_1 = 0$, $\tilde{\beta}_2 = \beta_1 + \beta_2$, and $\tilde{\beta}_3 = \beta_1 + \beta_3$. There is no unique solution: $(\beta_1, \beta_2, \beta_3)$ is one valid set of coefficients, but $(\tilde{\beta}_1, \tilde{\beta}_2, \tilde{\beta}_3)$ is equivalent, and is also a valid solution.

Note that in the case of regression with a single regressor, this condition just requires that the regressor, X_i , has non-zero variance.

Assumption (IID Sampling). *Each observation is drawn independently and from the identical distribution as the other observations.*

This assumption allows us to apply the law of large numbers. If each observation is drawn independently, then sample means will converge to population means. This will cause the OLS estimator to converge to its population counterpart, the best linear predictor.

Assumption (Orthogonality Condition). *The error term, ε_i , has zero mean $\mathbb{E}[\varepsilon_i] = 0$, and is uncorrelated with the regressors, $\text{Cov}(X_{k,i}, \varepsilon_i) = 0$ for each k .*

This assumption ensures that the best linear predictor is also the causal effect that we are interested in. In practice, we spend the most time thinking about whether the orthogonality condition holds. This assumption is sometimes also called an “exogeneity” assumption.

Proof of Consistency

With these three assumptions, we can show that OLS is consistent.

Theorem. *Suppose that the regressors are not perfectly collinear, each observation is sampled iid, and the orthogonality condition holds ($\mathbb{E}[\varepsilon_i] = 0$ and $\text{Cov}(X_{k,i}, \varepsilon_i) = 0$ for each k). Then, the OLS estimator is consistent:*

$$\begin{aligned} \hat{\alpha} &\xrightarrow{p} \alpha \\ \hat{\beta}_k &\xrightarrow{p} \beta_k \text{ for all } k. \end{aligned}$$

We'll provide a proof of consistency for the single regressor case: the proof for multiple regressors is similar. This proof will help make clear why we need each assumption.

First, note that last time, we showed that

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \cdot \bar{X}$$

We did not emphasize it at the time, but this required the “full rank” assumption. In the single regressor case, this assumption just means that we need $\text{Var}(X_i)$ to be non-zero. More generally, the full rank assumption ensures that there is a unique solution to the least-squares problem. If the full rank assumption holds, then each of the orthogonality conditions will be linearly independent, and we will have enough equations to pin down all of the unknown parameters.

Second, we will use the fact that each observation, (X_i, Y_i) , is drawn iid to show that $\hat{\beta}$ and $\hat{\alpha}$ converge to their population counterparts, β^{BLP} and α^{BLP} . Note that the sample covariance can be written as a function of sample means:

$$\begin{aligned}\widehat{\text{Cov}}(X_i, Y_i) &= \frac{1}{N} \sum_{i=1}^N (X_i Y_i) - \left(\frac{1}{N} \sum_{i=1}^N X_i \right) \cdot \left(\frac{1}{N} \sum_{i=1}^N Y_i \right) \\ &= \overline{XY} - \bar{X} \cdot \bar{Y}\end{aligned}$$

By the law of large numbers, since each observation is drawn iid, we have:

$$\begin{aligned}\overline{XY} &\xrightarrow{p} \mathbb{E}[X_i Y_i] \\ \bar{X} &\xrightarrow{p} \mathbb{E}[X_i] \\ \bar{Y} &\xrightarrow{p} \mathbb{E}[Y_i]\end{aligned}$$

Then, by Slutsky's Theorem, we can show that since these sample means all converge to their population counterparts, the sample covariance converges to the population covariance:

$$\begin{aligned}\bar{X} \cdot \bar{Y} &\xrightarrow{p} \mathbb{E}[X_i] \cdot \mathbb{E}[Y_i] \\ \overline{XY} - \bar{X} \cdot \bar{Y} &\xrightarrow{p} \mathbb{E}[X_i Y_i] - \mathbb{E}[X_i] \cdot \mathbb{E}[Y_i] \\ \implies \widehat{\text{Cov}}(X_i, Y_i) &\xrightarrow{p} \text{Cov}(X_i, Y_i)\end{aligned}$$

We can use equivalent arguments to then show that the sample variance converges to the population variance, $\widehat{\text{Var}}(X_i) \xrightarrow{p} \text{Var}(X_i)$. Finally, one more application of Slutsky's Theorem shows that $\hat{\beta} \xrightarrow{p} \beta^{BLP}$, and similarly $\hat{\alpha} \xrightarrow{p} \alpha^{BLP}$.

$$\begin{aligned}\hat{\beta} &= \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)} \xrightarrow{p} \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)} = \beta^{BLP} \\ \hat{\alpha} &= \bar{Y} - \hat{\beta} \cdot \bar{X} \xrightarrow{p} \mathbb{E}[Y_i] - \beta^{BLP} \cdot \mathbb{E}[X_i] = \alpha^{BLP}\end{aligned}$$

Thus, the fact that observations are drawn iid allows us to prove that our least-squares estimates converge to their population counterparts.

Finally, we will use the orthogonality condition to show that the best linear predictor, β^{BLP} , is equal to

the β in our model. We have:

$$\begin{aligned}\beta^{BLP} &= \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)} \\ &= \frac{\text{Cov}(X_i, \alpha + \beta \cdot X_i + \varepsilon_i)}{\text{Var}(X_i)} \\ &= \frac{\beta \cdot \text{Cov}(X_i, X_i) + \text{Cov}(X_i, \varepsilon_i)}{\text{Var}(X_i)} \\ &= \frac{\beta \cdot \text{Var}(X_i) + 0}{\text{Var}(X_i)} \\ &= \beta\end{aligned}$$

where we used the orthogonality condition $\text{Cov}(X_i, \varepsilon_i) = 0$. We can similarly show $\alpha^{BLP} = \alpha$:

$$\begin{aligned}\alpha^{BLP} &= \mathbb{E}[Y_i] - \beta^{BLP} \cdot \mathbb{E}[X_i] \\ &= \mathbb{E}[\alpha + \beta \cdot X_i + \varepsilon_i] - \beta^{BLP} \cdot \mathbb{E}[X_i] \\ &= \alpha + \beta \cdot \mathbb{E}[X_i] + \mathbb{E}[\varepsilon_i] - \beta^{BLP} \cdot \mathbb{E}[X_i] \\ &= \alpha + \mathbb{E}[\varepsilon_i] + (\beta - \beta^{BLP}) \cdot \mathbb{E}[X_i] \\ &= \alpha + 0 + 0 \cdot \mathbb{E}[X_i] \\ &= \alpha\end{aligned}$$

where we used $\mathbb{E}[\varepsilon_i] = 0$, plus our earlier result that $\beta^{BLP} = \beta$.

To summarize our proof:

- The full rank condition ensures that $\hat{\beta}$ exists and is unique.
- The iid condition ensures that least-squares in the sample converges to the best linear predictor in the population, $\hat{\beta} \xrightarrow{p} \beta^{BLP}$.
- The orthogonality condition ensures that the best linear predictor is the same as the effect we are interested in, $\beta^{BLP} = \beta$.

Unbiasedness of OLS

We have already seen that, under appropriate assumptions, OLS is a consistent estimator of the true causal effect. Consistency is a very desirable property for an estimator to have. Another desirable property is unbiasedness. We will now show that OLS is unbiased under similar assumptions: the only difference is that we have to somewhat strengthen our orthogonality assumption. We will first discuss this stronger assumption, and then we will show that OLS is unbiased.

Assumptions for Unbiasedness

To prove unbiasedness of OLS, we will need to replace the orthogonality condition with a stronger assumption: the zero conditional mean assumption.

Assumption (Zero Conditional Mean). *The error term, ε_i , has expected value of zero conditional on the regressors, $\mathbb{E}[\varepsilon_i | X_{1,i}, \dots, X_{K,i}] = 0$.*

This assumption is another version of exogeneity, but it is stronger than the earlier orthogonality condition, $\text{Cov}(X_{k,i}, \varepsilon_i) = 0$. In general, the zero conditional mean assumption implies orthogonality, but not

the other way around. In fact, if $\mathbb{E}[\varepsilon_i] = 0$, then we can connect independence, zero conditional mean, and orthogonality (also known as uncorrelatedness):

$$X_i \text{ and } \varepsilon_i \text{ independent} \implies \underbrace{\mathbb{E}[\varepsilon_i | X_i]}_{\text{Zero Conditional Mean}} \implies \underbrace{\text{Cov}(X_i, \varepsilon_i) = 0}_{\text{Orthogonality/Uncorrelatedness}}$$

That is, independence of the regressor and the error term implies zero conditional mean, and zero conditional mean implies orthogonality. We can think of independence as saying “there is no relationship between ε_i and X_i ,” the zero conditional mean condition as saying “there is no relationship between the mean of ε_i and X_i ,” and orthogonality as saying “there is no linear relationship between ε_i and X_i .”

This gets at the key difference between our earlier orthogonality condition and the zero conditional mean assumption. Orthogonality requires that the error term is uncorrelated with the regressor: this means that the linear relationship between the error and regressor has a slope of zero, but there could be a nonlinear relationship between the error and regressor. This connects to a very subtle point: by requiring the error term to have zero mean for every value of X_i , the zero conditional mean assumption imposes that the true causal model is linear. In contrast, the orthogonality condition does not require the true model to be linear: if the true model is nonlinear, but the orthogonality condition holds, then OLS will consistently estimate the β that provides the best linear approximation to the true causal model.

Exercise 3. Suppose that $\mathbb{E}[\varepsilon_i | X_i] = 0$. Show that this implies $\mathbb{E}[X_i \varepsilon_i] = 0$. *Hint: Use the law of iterated expectations to note that $\mathbb{E}[X_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[X_i \varepsilon_i | X_i]]$.*

Exercise 4. Why does $\mathbb{E}[X_i \varepsilon_i] = 0$ imply that $\text{Cov}(X_i, \varepsilon_i) = 0$?

Proof of Unbiasedness

With our earlier assumption of no perfect collinearity and iid sampling, along with this zero conditional mean assumption, we can show that OLS is unbiased.

Theorem. Suppose that the regressors are not perfectly collinear, each observation is sampled iid, and the error term has zero mean conditional on the regressors ($\mathbb{E}[\varepsilon_i | X_{1,i}, \dots, X_{K,i}] = 0$). Then, the OLS estimator is unbiased:

$$\begin{aligned}\mathbb{E}[\hat{\alpha}] &= \alpha \\ \mathbb{E}[\hat{\beta}_k] &= \beta_k \text{ for all } k.\end{aligned}$$

We'll now provide a proof of unbiasedness for the single variable case, focusing on the slope parameter, β . We can write the least-squares estimator as:

$$\begin{aligned}\hat{\beta} &= \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)} \\ &= \frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) \cdot Y_i\end{aligned}$$

We can then take the expectation and rearrange terms:

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}\left[\frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) \cdot Y_i\right] \\ &= \mathbb{E}\left[\frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) \cdot (\alpha + \beta \cdot X_i + \varepsilon_i)\right]\end{aligned}$$

From here, we can use the law of iterated expectations, noting that the unconditional expected value is equal to the expected value of the conditional expectation. We can condition on X , which represents all

draws of the regressor $(X_1, X_2, \dots, X_N)$. This lets us move functions of X outside of the expectation:

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E} \left[\mathbb{E} \left[\frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) \cdot (\alpha + \beta \cdot X_i + \varepsilon_i) \mid X \right] \right] \\ &= \mathbb{E} \left[\frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) (\alpha + \beta \cdot X_i + \mathbb{E}[\varepsilon_i \mid X]) \right]\end{aligned}$$

Finally, note that since the data is drawn iid, $\mathbb{E}[\varepsilon_i \mid X] = \mathbb{E}[\varepsilon_i \mid X_i]$, since the distribution of ε_i will not depend on any other observation's draw of X_i . Since we have assumed that $\mathbb{E}[\varepsilon_i \mid X_i] = 0$, we then have:

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E} \left[\frac{1}{\widehat{\text{Var}}(X_i)} \cdot \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) (\alpha + \beta \cdot X_i + 0) \right] \\ &= \mathbb{E} \left[\frac{\widehat{\text{Cov}}(X_i, \alpha + \beta \cdot X_i)}{\widehat{\text{Var}}(X_i)} \right] \\ &= \mathbb{E} \left[\frac{\beta \cdot \widehat{\text{Cov}}(X_i, X_i)}{\widehat{\text{Var}}(X_i)} \right] \\ &= \beta \cdot \mathbb{E} \left[\frac{\widehat{\text{Var}}(X_i)}{\widehat{\text{Var}}(X_i)} \right] \\ &= \beta\end{aligned}$$

Exercise 5. What is the difference between unbiasedness and consistency? When is unbiasedness more useful? When is consistency more useful?